Here is a summary of the landscape regarding public fluoroscopic datasets, those specific to the spine, and the major frameworks used to realistically simulate fluoroscopic noise and physics.

1. Public Fluoroscopic X-Ray Datasets

Because true fluoroscopy involves live intraoperative video of patients, it is heavily restricted by HIPAA and global privacy laws. Consequently, most public datasets use cadavers or synthetic bone phantoms.

- **Veriserum (2025):** Currently the largest open-source *dual-plane* fluoroscopic dataset. It contains ~110,000 X-ray images of knee implant phantoms moving through activities of daily living, complete with calibration tools and ground-truth poses. Hosted by ETH Zürich.
- **DeepFluoro (IPCAI 2020):** A highly cited single-plane fluoroscopy dataset providing accurate 2D/3D registration and pose labels specifically for **hip anatomy**. Created by researchers at Johns Hopkins.
- **CAMS-Knee & Stan Datasets:** High-quality fluoroscopic datasets focused on total knee arthroplasty (knee implants) that provide precise 3D kinematic tracking alongside the 2D fluoroscopic video.

2. Fluoroscopic Datasets Containing the Spine

There are virtually zero open-access, direct-download datasets of real human spine fluoroscopy. You must either request access from authors directly or rely on datasets that project 3D CTs into 2D to simulate fluoroscopy.

- **CM Spine Dataset (2024):** Contains 208 real intraoperative G-arm images of the lumbar spine used to train a Graph Neural Network (GNN) for vertebral detection. (Requires directly emailing the authors at the Chinese Academy of Sciences for access).
- **LevelCheck Clinical Data:** The Johns Hopkins LevelCheck project has a vast repository of real intraoperative spine fluoroscopy paired with preoperative CTs. It is strictly available via a formal Data Use Agreement with the Principal Investigator.
- **CTSpine1K / X23D (Simulated):** The most common workaround for spine researchers. This dat
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3. Replicating Fluoroscopic & X-Ray Noise Synthetically

Because real data is scarce, the field relies heavily on simulating fluoroscopy from CT scans. Simply taking a 2D ray-cast of a CT scan looks far too clean to train an AI model. The following papers and frameworks are the gold standard for accurately modeling the physical noise (Poisson quantum mottle, electronic noise, and scatter) of a real C-arm:

- **DeepDRR (MICCAI 2018 - *A Catalyst for Machine Learning in Fluoroscopy-guided Procedures*):** This is the most famous framework for this exact problem. DeepDRR explicitly combines analytic X-ray physics (Beer-Lambert attenuation) with machine learning for scatter estimation and signal-dependent noise injection. It specifically models the noise characteristics of a flat-panel C-arm detector so models trained on synthetic DeepDRR images can seamlessly run on real operating room machines.
- **DiffDRR & ProST:** These are modern, differentiable extensions of the physics-based simulation principles established by DeepDRR. Because they are differentiable, you can plug them directly into a neural network training loop to actively optimize the noise and scatter parameters during training.
- **Real-Time Medical Video Denoising with Deep Learning (NIH/Yale):** Rather than generating noise, this paper tackles the inverse problem. The authors trained deep convolutional autoencoders explicitly on fluoroscopic X-ray images corrupted with Gaussian, speckle, and salt-and-pepper noise to create a real-time (sub-millisecond) denoising filter that matches the physical characteristics of live medical angiograms.